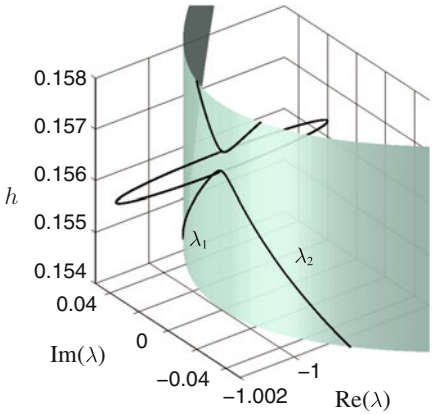


Fig. 4.3 Eigenvalues of the simple multiple timestepping method. The eigenvalues coil around the surface of the cylinder $|\lambda| = 1$ except at exceptional points near $k\pi/\Omega$, where $k \in \mathbb{Z}$. Near these points of resonance, exponential instabilities are present

and the norm of the solution will oscillate and grow linearly with n as $n \to \infty$. This is the simplest type of *linear resonance* present in multiple timestepping. The resonance at $h = \pi/\Omega$ only gives rise to a relatively weak 'linear' instability, but what is more serious is that in the vicinity of $h = \pi/\Omega$ the eigenvalues actually leave the unit circle. We can see this by expanding $\mu(h) = 2\cos(h\Omega) - \frac{h}{\Omega}\sin(h\Omega)$ in a Taylor expansion around $h = \pi/\Omega$:

$$\mu(h) = -2 + \varepsilon, \qquad \varepsilon = \frac{\pi}{\Omega}(h - \pi/\Omega).$$

Both eigenvalues satisfy

$$\lambda + \lambda^{-1} = -2 + \varepsilon + \mathcal{O}(\varepsilon^2).$$

In case $\varepsilon > 0$, i.e. *after* $h = \pi/\Omega$, we have complex eigenvalues and we already know this case to be stable. For $\varepsilon < 0$, however, we have

$$\lambda = -1 \pm (|\varepsilon|/2)^{1/2} + \mathcal{O}(\varepsilon).$$

Here both eigenvalues are real and one is outside the unit circle. There is thus a small interval of *exponential* instability directly below $h = \pi/\Omega$.

We can easily solve for the two eigenvalues as functions of h and graph them as points in three dimensions: $x = \mathrm{Re}(\lambda)$, $y = \mathrm{Im}(\lambda)$, $z = h$ (Fig. 4.3). Such resonances will also occur at integer multiples of π/Ω, and we can therefore expect many small intervals in the stepsize where the multiple timestepping method is unstable. When a large system involves a combination of many different interactions, the intervals of instability would expect to proliferate from all the different natural frequencies.

A linear resonance is a rational relationship between the period of the harmonic oscillator and the timestep used in simulation. If we substitute a nonlinear oscillator